Causal Inference Project

Group members: Léo Bouchand and Nicolas Le Jeune

Context & project

In August 2022, the United States passed two industrial policy laws: the CHIPS and Science Act, which subsidizes domestic semiconductor production, and the Inflation Reduction Act (IRA), which provides large tax credits for clean energy and its manufacturing supply chain (batteries, solar, electric vehicles).

Since then, spending on factory construction in the United States has risen sharply, especially in the computer, electronic and electrical sector. However, this increase happened at the same time as other trends, such as reshoring after COVID, rising construction costs and higher interest rates.

Our goal is to estimate how much of this increase in industrial investment is caused by the IRA and the CHIPS Act. We plan to use a difference-in-differences approach, comparing sectors targeted by these laws with non-targeted sectors before and after August 2022, as well as a synthetic control, building a counterfactual United States from other OECD countries.

Data

Our main outcome is monthly private construction spending from the US Census Bureau (2010-2026), broken down by building type and by manufacturing subsector. We complement it with FRED series (construction material prices, industrial production and employment, interest rates), quarterly OECD data on non-residential construction investment for the United States and about twenty other countries, and data from the Clean Investment Monitor (Rhodium Group and MIT) on clean technology investment by technology and by state.